

a promising new way of powering everything—from homes and cars to portable devices like cellphones and laptop computers. Their major advantage—the prospect of eliminating emissions of greenhouse gases and other pollutants—has been outweighed by their very high cost. Researchers are therefore trying to find ways to make the devices less expensive.

Researchers at the Massachusetts Institute of Technology have found a method that has the potential to dramatically increase the efficiency of electrodes in one type of fuel cells which uses methanol instead of hydrogen as its fuel. Researchers believe that this battery can replace the expensive batteries. Since these electrodes are made of platinum, increasing their efficiency means less use of the expensive metal to produce power.

By creating tiny stair-steps to the surface instead of leaving it smooth, the electrode's ability to catalyse oxidation of the fuel and thus produce electric current was approximately doubled during the experiments. Researchers believe that further development of these surface structures could result in producing more electric current.

Multi-touch Magic Mouse

Apple has introduced the new wireless Magic Mouse with multi-touch technology. Pioneered on iPhone, iPod touch and Mac notebook trackpads, multi-touch allows customers to navigate using intuitive finger gestures. Instead of mechanical buttons, scroll wheels or scroll balls, the entire top of the Magic Mouse is a seamless multi-touch surface.

Magic Mouse comes with every new iMac. You can also add it to any Bluetooth-enabled Mac for a multi-touch makeover.



Apple Magic Mouse

The mouse uses powerful laser tracking that's far more sensitive and responsive on more surfaces than traditional optical tracking. It tracks with precision on nearly every surface without the need for a mousepad.

Flash player for smartphones

Adobe is coming out with a Flash Player 10.1 software for smartphones, smartbooks, netbooks, PCs and other Internet-connected devices. A public developer beta of the browser-based runtime is expected to be available for Windows Mobile, Palm webOS and desktop operating systems. Public betas for Google Android and Symbian OS will be available in 2010.

The browser-based runtime leverages the power of the graphics processing unit (GPU) for accelerated video and graphics while conserving battery life and minimising resource utilisation.

Plastic Logic for e-reader

After spending nine years in development and tens of millions of dollars in funding, Plastic Logic is finally ready to debut the first electronic reader made of plastic electronics. The long-awaited e-reader—the QUE proReader—will be officially launched in January 2010 at the Consumer Electronics Show. It is about the size of a 21.6 × 28cm notepad.

The QUE proReader claims to feature the largest touchscreen in the industry and a touchscreen user interface. It will support PDF, Word, PowerPoint and Excel formats, and also features tools for managing content.

Solar-powered touchscreen phone

LG Electronics has launched its new touchscreen phone—LG GD510—that can be recharged utilising the power of the sun. A 10-minute exposure to sunlight provides up to 2



Plastic Logic QUE proReader

minutes 15 seconds of talk time or up to 180 minutes of standby time. Features like eco-tree and eco-calculator under the phone's CO₂ menu allow users to track how much they have cut down on CO₂ emissions by using the solar cell battery.

When the phone is fully charged, it beeps and displays a message on the screen, reminding users to unplug the charger and save energy. Its packaging is also made from recycled paper with soy ink printing.

LG GD510 features a WQVGA wide screen surrounded by a narrow 4.8mm bezel and smooth metal body with just a single button on the front. It also has all multimedia features like 3MP camera, MP3 player and Web browser.



LG GD510

90nm embedded flash technology

STMicroelectronics has achieved a breakthrough in microcontrollers units (MCUs) based on the ARM Cortex-M series of processor cores. It features embedded flash of 90nm process technology. These MCUs will deliver faster operation, increased peripheral integration and save power. They will also provide the highest on-chip static RAM (SRAM) and non-volatile memory densities.